



17.5

Section: 4 Quiz: 2

Marks: 20

Time: 30 minutes

1. Suppose that your demand schedule for DVDs is as follows:

Price	Quantity Demanded (Income = \$10,000)	Quantity Demanded (Income = \$12,000)
\$ 8	40 DVDs	50 DVDs
10	32	45
12	24	30
14	16	20
16	8	12

Use the midpoint method to calculate your price elasticity of demand as the price of DVDs increases from \$10 to \$12 if (i) your income is \$10,000 and (ii) your income is \$12,000. (Indicate the type of goods in each case) [5+5=10 marks]

① Price elasticity demand

$$= \left| \frac{\frac{24 - 32}{(24 + 32)/2}}{\frac{12 - 10}{10 + 12}} \right|$$

$$= \left| \frac{\frac{-8}{28}}{\frac{2}{11}} \right| = |-1.571| = 1.571 > 1$$

as in PED > 1 so, it is elastic and normal good

$$\text{② PED} = \left| \frac{\frac{30 - 45}{(30 + 45)/2}}{\frac{12 - 10}{10 + 12}} \right| = \left| \frac{\frac{-15}{37.5}}{\frac{2}{11}} \right| = |-2.2| = 2.2 > 1$$

So, it is elastic and normal good

b. Calculate your income elasticity of demand as your income increases from \$10,000 to \$12,000 if the price is \$14 (Indicate the type of good) [5 marks]

$$IED = \frac{20 - 16}{[(20 + 16)/2]}$$

$$= \frac{120,000 - 10,000}{10,000 + 120,000}$$

$$= \frac{2}{4} = \frac{1}{2}$$

$$= \frac{2000}{11000} = 1.22 > 1$$

AS $IED > 1$
it is normal
good and
elastic.

2. You have the following information about good X and good Y:

- Income elasticity of demand for good X: -3
- Cross-price elasticity of demand for good X with respect to the price of good Y: 2

Would an increase in income and a decrease in the price of good Y unambiguously decrease the demand for good X? Why or why not? [5 marks]

X is an inferior good, so if I increase the income and decrease the price of Y (which is a substitute of X) the demand will be increased. When the price of substitute decreases the demand will be increased of product X. because, we know that (substitute of X is Y)